

PS200C: Quantitative Methods

In-class Activity 9: Regression Discontinuity Designs

Name: _____

Setup for Q1. In the United States, Medicare provides nearly universal health insurance to people aged 65 and older. At age 65 a person becomes eligible for Medicare regardless of their previous insurance status. A researcher wants to estimate the effect of Medicare on hospital utilization, using age as the running variable.

Question 1: Identify the pieces. Write down what the running variable X_i , the cutoff c , the treatment D_i , and a candidate outcome Y_i are in this design. Then state, in one sentence, what the continuity assumption requires here.

Question 2: What else changes at the cutoff? The biggest threat to RD is another treatment at the same threshold (“bundled treatments”). Name *at least one* other thing that changes at age 65 that could plausibly affect hospital utilization, and explain in a sentence why it is a problem for the RD interpretation.

Question 3: Reading a balance plot. Suppose you run an RD on Brazilian municipalities at the e-voting threshold (40,500 registered voters) and produce balance plots like the one we saw in lecture. For each finding below, say in one sentence whether it *supports* the design, *threatens* the design, or is *ambiguous*, and why.

- (a) Pre-1998 municipal income shows no jump at the cutoff.
- (b) Pre-1998 literacy rate shows a small but statistically significant jump at the cutoff.
- (c) *Post-1998* federal social spending shows a jump at the cutoff.

Question 4: Sharp vs. fuzzy. For each scenario below, say whether you'd treat it as a **sharp** or **fuzzy** RD, and – if fuzzy – what the estimand becomes (and for whom).

- (a) French municipal elections: cities with population $\geq 3,500$ use proportional representation; below 3,500 they use first-past-the-post. Outcome: number of effective parties.
- (b) A scholarship program grants automatic offers to students scoring above an SAT threshold; students just below the threshold can still be considered, but the probability of an offer drops sharply. Outcome: college completion.